Advanced Intelligent Anemia Detection System

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***Abstract*—This paper explores non-invasive anemia detection using advanced machine learning models applied to images of hands, palms, and fingernails. Anemia remains a critical global health issue, especially in underserved regions where traditional diagnostics may be inaccessible. The study evaluates methods such as Convolutional Neural Networks and ensemble techniques for their effectiveness in image-based diagnosis. Emphasis is placed on smartphone integration to enable accessible, real-time analysis. Findings highlight strong diagnostic potential while noting the need for larger, diverse datasets and more robust models to improve accuracy and real-world adaptability.**

Keywords—Anemia detection, non-invasive diagnosis, image analysis, CNN, machine learning, LLM chatbot, mobile health, Flutter, FAISS, segmentation.

I. Introduction

Anemia is a common health concern, particularly in underserved communities, marked by low levels of hemoglobin or red blood cells, and is typically identified through invasive blood tests [1], [26], [38]. This project presents a mobile application powered by AI that identifies anemia without invasive methods by examining images of the conjunctiva, fingernails, and palms through the use of machine learning models such as CNNs [2], [5], [6], [7], [8], [9], [10], [17], [18], [21], [23], [25]. The application features a chatbot powered by a Large Language Model (LLM) that provides real-time health feedback, tailored guidance, and medical recommendations [13]. The goal is to boost early detection, improve healthcare accessibility, and lower diagnostic costs by utilizing an intuitive interface that demands minimal training for users [3], [4], [14], [15], [16], [36]. The solution accommodates a wide range of skin tones, lighting situations, and device functionalities, featuring real-time processing and secure data management. [4], [16], [26], [27], [28], [29], [30]. Utilizing smartphone technology and on- device AI, the system facilitates rapid and precise screening, along with the possibility of integration with telemedicine services. [12], [16], [36]. This project emphasizes its significance in public health, providing a scalable tool for anemia screening among various populations while maintaining compliance with data privacy standards [19], [37]. This project stems from a collective team endeavor that encompasses data gathering, model creation, application design, implementation, and thorough validation [1], [2], [5],
[13], [19].

The remainder of this paper is organized as follows: Section II reviews related work. Section III describes the proposed system architecture. Section IV details the datasets and preprocessing steps. Section V presents model integration and the mobile application design. Section VI discusses model interpretability. Section VII addresses patient data privacy. Section VIII highlights deployment challenges. Section IX presents limitations and future work. Finally, Section X concludes the paper.

II. Related Work

Prior research in non-invasive anemia detection has extensively explored the application of machine learning (ML) algorithms to visual data, particularly images of the conjunctiva, fingernails, and palms [2], [5], [18], [19], [25]. Traditional diagnostic approaches—while accurate—depend on invasive blood tests and laboratory infrastructure, posing challenges in terms of cost, discomfort, and inaccessibility, especially in low-resource or remote environments [1], [4], [16], [26], [28], [36]. To address these limitations, a variety of ML models have been employed to classify anemia based on image analysis [2], [3], [9], [19], [23], [31], [32], [33], [34], [35]. Among these, Convolutional Neural Networks (CNNs) have demonstrated superior performance, achieving accuracies as high as 99.12%, particularly when analyzing conjunctiva images, which strongly correlate with hemoglobin levels [6], [14], [19], [20], [21], [24], [30]. Other models like Support Vector Machines (SVMs), k-Nearest Neighbors (k-NN), Decision Trees, and Naïve Bayes have shown promise with smaller datasets or in low-compute environments, though typically with slightly lower performance [19], [20], [21], [22], [23], [24], [31], [35]. Several studies also incorporated these models into smartphone applications, allowing users to detect anemia through real-time image capture and analysis using mobile cameras [3], [4], [11], [12], [16], [36], [37], [38], [39], [40].

These mobile solutions have achieved up to 92.5% accuracy and offer potential for large-scale deployment due to their portability and ease of use [4], [11], [28], [36]. However, prior systems face key limitations, including lack of standardized lighting and image capture conditions, limited dataset diversity, and challenges in generalizing across different populations and devices [19], [25], [33]. These constraints highlight the need for scalable, accessible, and intelligent diagnostic systems— such as our proposed solution, which combines CNN-based processing with LLM-powered user interaction [13], [21], [34]. A summary of

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performance metrics from notable studies is provided below in Table I.

A. Summary of Related Models

TABLE I. RELATED WORK

Model	Metrics				
	Model	Accuracy	Sensitivity	Precision	F1 Score
Palm	CNN [23]	99.92	99.98	99.79	99.89
	DT [20]	95.62	97.92	98.60	98.01
Fingernails	CNN [21]	98.33	97.44	97.64	97.54
	KNN [21]	90.26	85	85	84
Conjunctiva	DT [21]	97.32	98.49	93.67	96.02
	ViT [3]	84.08	83.5	83.3	83.3

III. PROPOSED SYSTEM ARCHITECTURE

The proposed system is a unified AI-powered platform designed for fast, non-invasive anemia screening, supported by an interactive medical assistant. The architecture consists of an image-based diagnostic pipeline and an optional LLM-based chatbot for interpretation, education, and guidance, as shown in Fig. 1.

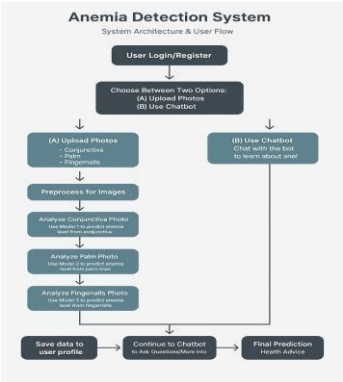

Fig. 1 Anemia detection system from system architecture and user flow

A. User Flow and Modalities

After secure login, the user follows a guided workflow to capture or upload three key images: palpebral conjunctiva, palm, and fingernails. The system is designed to reduce common mobile capture issues such as blur, poor illumination, and inaccurate framing. Once images are provided, the pipeline processes each modality independently and produces modality-level confidence estimates. These outputs are then combined through decision fusion to produce a final screening result. The user may then optionally consult the chatbot for explanation and next-step guidance.

B. ROI Segmentation and Standardization

To improve robustness under real-world capture conditions, the pipeline begins with ROI segmentation to localize clinically informative regions. For each modality, segmentation isolates the relevant area (inner eyelid region for conjunctiva, central palm region, and nail-bed region for fingernails), reducing background noise and improving consistency across devices and environments. The segmented ROI is then standardized (e.g., resized to a fixed input size such as 224×224 and normalized) before classification. Representative examples of raw images and segmented ROIs are shown in Fig. 2.

Fig. 2 ROI segmentation examples

C. Modality-Specific Classification

Following segmentation, modality-specific classifiers estimate anemia likelihood from the standardized ROI images. Each classifier produces a confidence score for its modality. This design allows the system to exploit complementary cues across conjunctiva, nail-bed, and palm appearance while mitigating the impact of noise or weakness in any single modality.

D. Decision Fusion and Final Output

A decision-level fusion mechanism aggregates modality-level predictions into a single final screening output (e.g., no anemia, low likelihood, or high likelihood of anemia). Fusion improves stability compared to single-modality inference and reduces sensitivity to occasional modality failures. In uncertain or low-quality cases, the system may encourage recapture or place greater emphasis on the most reliable available modality outputs.

E. Optional Chatbot Guidance with Doctor and Patient Modes

After receiving the fused screening result, the user may optionally interact with an integrated chatbot. The chatbot supports two modes: Doctor Mode, which provides more technical explanations and clinically oriented reasoning, and Patient Mode, which presents simplified, user-friendly explanations and actionable advice. This dual-mode design improves accessibility and helps users understand results and next steps.

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IV. Dataset and Preprocessing

This study leverages multimodal datasets across three key domains—image classification, segmentation, and language modeling—to build a robust anemia detection system powered by deep learning and conversational AI.

A. Classification Datasets

We use labeled images of the conjunctiva, fingernails, and palms, each categorized as anemic or non-anemic. Initial class imbalance (e.g., 2052 vs. 1297 samples in fingernail data) was addressed through data augmentation using Augmentations (flips, rotations, resizing to 224×224). After balancing, each class included 2000+ images, ensuring fairness and improved model generalization.

B. Segmentation Datasets

To localize disease-relevant regions, we used three segmentation datasets:

- Conjunctiva (180 images): segment the Inner Eyelid.
- Palm (420 images):bounding boxes for central palm zones.
- Nails (52 images): Binary masks isolating nail beds.

These regions were segmented using U-Net and DeepLabV3 architectures, enhancing downstream classification by removing background noise.

C. LLM Chatbot Datasets

To train the AI health assistant, we adopted the "AI Medical Chatbot" dataset (256K+ Q&A pairs). From this, 863 anemia-related interactions were extracted and analyzed. Patient query lengths and doctor response patterns informed prompt tuning and language model constraints.

V. Model Integration and Application Design

This section elaborates on the core technological components of the proposed system, detailing the machine learning models for anemia classification, the deployment of a Large Language Model (LLM)-powered chatbot for medical conversations, and the design and implementation of the Flutter-based mobile application.

A. Classification Models and Training

To accurately classify anemia based on visual inputs from the palm, fingernail, and conjunctiva, a suite of machine learning models was trained and evaluated. Each biological region was associated with a set of models optimized for high F1-score performance under varying lighting and image conditions. Table II provides proposed work performance metrics.

TABLE II. Proposed Work

Model	Metrics				
	Model	Accuracy	Sensitivity	Precision	F1 Score
Palm	RF	99.54%	100%	99%	99%
Fingernails	RF	100%	100%	100%	100%
Conjunctiva	XGBoost	99%	99%	99%	99%

B. LLM-Based Medical Chatbot Integration

To complement the diagnostic pipeline, a robust LLM powered medical chatbot was integrated to simulate nuanced doctor–patient interactions and deliver context-aware, real time clinical support. The chatbot is built upon the DeepSeek-R1-Distill-Qwen-7B architecture, a 7-billion parameter model distilled from biomedical corpora, ensuring both domain specificity and inference efficiency. Leveraging autoregressive decoding, the system enables natural turn taking and dynamic dialogue generation, offering medically accurate responses aligned with professional standards. To optimize performance for edge deployment, the model employs 4-bit quantization and supports distributed inference via multi-GPU configurations, enabling low-latency responses even under constrained resources. A hybrid retrieval mechanism—integrating FAISS-based dense vector search with metadata-aware filtering—enriches the chatbot's contextual understanding, allowing precise and semantically relevant responses for both exact and fuzzy queries. The interaction is tailored through dual conversational modes: Doctor Mode, which delivers expert-level explanations with clinical reasoning and technical detail; and Patient Mode, which provides user-friendly summaries suitable for non-specialists. The chatbot supports a wide range of tasks, including symptom clarification, diagnostic explanation, and personalized post-detection guidance—thereby enhancing accessibility, promoting early intervention, and aligning seamlessly with the mobile application's broader goal of democratizing healthcare through intelligent automation.

C. Flutter Application Design

The cross-platform mobile application was developed using Flutter, enabling native-like performance and seamless compatibility across both Android and iOS platforms. The app supports the complete user journey—from account registration and authentication to AI-driven anemia detection and interactive medical consultation via chatbot. Key modules include a secure login/signup system integrated with Firebase Authentication; a dashboard that guides users to either the image-based diagnostic module or the LLM powered chatbot; an image upload interface supporting conjunctiva, fingernail, and palm submissions; a results screen displaying detection output along with severity that categorization (low, moderate, or high); and a chatbot interface enables real-time medical dialogue. Additionally, the app offers user profile management with editable personal details and photo upload. From a technical standpoint, AI/ML inference is handled via backend RESTful APIs, ensuring device-agnostic intelligence while maintaining responsiveness. The UI employs a responsive

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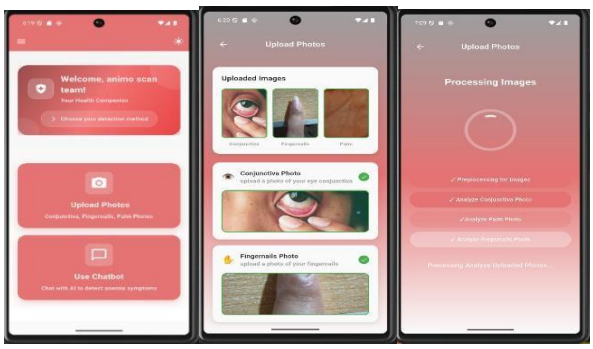
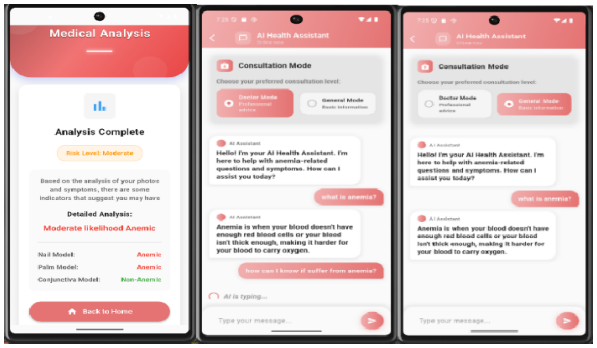
design paradigm, ensuring consistent user experience across varying screen sizes and devices. Deployment requires a backend server hosting trained AI models, Firebase services for authentication and analytics, and a FAISS-based vector index for efficient semantic retrieval in chatbot queries. The app emphasizes usability, scalability, and secure data flow, making it a practical solution for both clinical and community health applications. Figure 3 (a), (b), and (c) illustrate the interface for the proposed anemia detection system interface.

(a)



(b)



(c)

Fig. 3 The proposed anemia detection system interface

VI. Model Interpretability

Interpretability is ensured by the system by means of a rule-based aggregate of predictions from palm, fingernail, and conjunctiva classifiers. Additionally, separate severity levels are provided in order to facilitate understandability of the results. Therefore, despite the fact that there are no visual explanation tools available, such as Grad-CAM, the rationale remains to be understandable and easy to communicate. Understanding is further increased by an integrated LLM chatbot, which offers explanations that are user-friendly and relevant, and which assists interpretation at both the doctor and patient levels. This chatbot also gives support for interpretation.

VII. Data Privacy for Patients

Through Firebase Authentication and secure API-based inference, the system places a high priority on protecting the privacy of its users. After the processing is complete, diagnostic pictures are not saved but rather erased. However, chatbot discussions are not documented, despite the fact that basic user data is managed. The system adheres to robust data protection policies, which ensure safe transmission and limited data retention throughout the pipeline, despite the fact that it is not technically certified under either the General Data Protection Regulation (GDPR) or the Health Insurance Portability and Accountability Act (HIPAA).

VIII. Deployment Challenges

Both the scalability of the model and the reliability of the application programming interface (API) presented a number of challenges during the deployment process. Stability was limited as a result of the vulnerability that was based on Ngrok, and the hosting of the LLM chatbot required a significant amount of resources that were dedicated to information processing. The presence of these limitations has resulted in an urgent need for deployment choices that are not only more durable but also more cost-effective. Obtaining real-time performance was accomplished by the use of application programming interfaces (APIs) that were modular in nature and backend design that was efficient.

IX. Limitations and Future Work

A. *Limitations*

Although the proposed system demonstrates promising performance for non-invasive anemia screening, several limitations remain. First, image-based inference is sensitive to real-world capture variability, including illumination, camera quality, motion blur, and inconsistent framing, which can reduce reliability outside controlled settings. Second, the current evaluation is limited by dataset size and diversity across skin tones, age groups, and device types, which may affect generalization to broader populations. Third, the system currently provides screening-level outputs and should not be interpreted as a definitive diagnosis; clinical confirmation via laboratory testing is still required. Finally, while the chatbot supports explanation and guidance, its responses may still be affected by retrieval coverage and prompt constraints, requiring continuous monitoring and safety controls for medical use.

B. *Future Work*

Future work will focus on:

- Improve usability and accessibility by adding multilingual support and optional voice input/output.
- Conduct clinical validation in hospitals and clinics, incorporating clinician feedback to support real-world adoption.

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- Extend the diagnostic output beyond binary screening to predict severity (mild/moderate/severe) and, where validated data permits, provide preliminary anemia-type indicators.
- Enable optional integration with health information systems and large-scale screening programs to support deployment in remote and underserved areas.

X. CONCLUSIONS

For the purpose of this investigation, multimodal datasets are utilized. These datasets encompass three key domains: language modelling, segmentation, and picture classification techniques. Building a robust anemia detection system that is powered by deep learning and conversational artificial intelligence is made possible via the utilization of these datasets, which allow for the building of the system. The solution that is being given as a solution is a unified platform that is powered by artificial intelligence and is designed to deliver a rapid and non-invasive diagnosis of anemia in addition to an interactive health assistant. This platform is being supplied as a solution. By utilizing photographs of the palm, fingernail, and conjunctiva, the application system demonstrated a high degree of accuracy in the identification of anemia. This was accomplished by the use of photographs. When compared to the results of earlier study, this paper was a substantial improvement. Additionally, the application system features a user interface that is simple to comprehend and provides two modes: one mode is designed for the physician, and the other mode is designed for the patient.

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